

MATH 345 Skeleton Notes

Unit 5: Optimization and training

Section 5.3: Eigenvalues, eigenvectors, and diagonalization

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Basic definitions

Definition Let A be an $n \times n$ matrix. Suppose $\mathbf{v}$ is a *nonzero* vector in $\mathbb{R}^n$ and λ is a scalar (which could be 0) such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Then λ is an **eigenvalue** of A , and $\mathbf{v}$ an **eigenvector** of

A associated with the **eigenvalue** λ .

Activity

Check that

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

are eigenvectors for the matrix

$$A = \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix},$$

and find the corresponding eigenvalues.

Activity

Let

$$A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}$$

be a matrix. Find the eigenvalues and eigenvectors of A .

Definition Let $A = [a_{ij}]$ be an $n \times n$ matrix. Then

$$\det(\lambda I_n - A) = \det \begin{bmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{bmatrix}$$

is called the

characteristic polynomial

of A and is sometimes

denoted $c_A(\lambda)$. The equation

$$\det(\lambda I_n - A) = 0$$

is called the

characteristic equation

of A .

Theorem Let A be an $n \times n$ matrix. The eigenvalues of A are the *roots* of the characteristic polynomial of A .

Remark Some authors define the characteristic polynomial to be $\det(A - \lambda I_n)$ rather than $\det(\lambda I_n - A)$. The advantage of the definition chosen in these notes is that the characteristic polynomial is always

monic

, i.e. the coefficient associated with the highest power of λ is always equal to one. The zeros of the two polynomials are both the same, and as we saw in Activity act-7-1-find-eigenvalues, they are the eigenvalues of A .

Activity

Find the characteristic polynomial of A if

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}.$$

Note: Finding eigenvalues and eigenvectors of a matrix A

1. Determine the roots of the characteristic polynomial $c_A(\lambda) = \det(\lambda I_n - A)$. These are the *eigenvalues* of A .

2. For each eigenvalue λ , find all nontrivial solutions to the homogeneous system $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$. These are the *eigenvectors* of A associated with λ .

Definition A basic solution (recall Basic solutions to homogeneous linear equations) to the homogeneous equation $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$ will be called a

basic eigenvector associated with the eigenvalue λ .

Warning Do not make the common mistake of first row-reducing A to RREF and then finding the eigenvalues and eigenvectors of the RREF matrix. Row operations on the matrix A *do not* preserve the set of basic solutions. They only preserve solutions to the homogeneous equation $A\mathbf{v} = \mathbf{0}$.

If $\mathbf{v}$ is an eigenvector of a matrix A with eigenvalue λ , then any nonzero scalar multiple of $\mathbf{v}$ is also an eigenvector with the same eigenvalue λ . Indeed, if $\mathbf{w} = c\mathbf{v}$ for some scalar $c \neq 0$, then:

$$\begin{aligned} A\mathbf{w} &= A(c\mathbf{v}) \\ &= c(A\mathbf{v}) \quad (\text{by linearity of matrix multiplication}) \\ &= c(\lambda\mathbf{v}) \quad (\text{since } \mathbf{v} \text{ is an eigenvector with eigenvalue } \lambda) \\ &= \lambda(c\mathbf{v}) \\ &= \lambda\mathbf{w} \end{aligned}$$

Put differently, the entire line $\text{span}(\mathbf{v})$ is “invariant” under the map defined by A , with every vector in this line being stretched (or compressed) by the factor λ .

Fact Any non-zero linear combination of eigenvectors associated with the *same* eigenvalue λ is also an eigenvector with eigenvalue λ . Conversely, all eigenvectors associated with the eigenvalue λ can be written

uniquely as a linear combinations of the basic eigenvectors associated with the eigenvalue λ .

Activity

Find the eigenvalues and basic eigenvectors of the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Diagonalization

Definition An $n \times n$ matrix A is called

diagonalizable

if there exists an invertible $n \times n$ matrix P

so that $P^{-1}AP$ is a diagonal matrix (recall Triangular and Diagonal Matrices). The matrix P is called a

diagonalizing matrix

for A .

Note that if $P^{-1}AP$ is a diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_n$, then for each standard basis vector (recall Standard basis) $P^{-1}AP\mathbf{e}_i = \lambda_i\mathbf{e}_i$. Multiplying this equation by P on the left hand side and setting $\mathbf{v}_i = P\mathbf{e}_i$ we obtain that

$$A\mathbf{v} = P(P^{-1}AP\mathbf{e}_i) = P(\lambda_i\mathbf{e}_i) = \lambda_i\mathbf{v}_i.$$

Thus $\mathbf{v}_i$ is an eigenvector of the matrix A associated with the eigenvalue

λ_i . Conversely,

Activity

Suppose

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 0 \end{bmatrix}.$$

Task Compute $P^{-1}AP$ where

$$P = \begin{bmatrix} -1/2 & -1 \\ 1 & 1 \end{bmatrix}.$$

You may use the fact that

$$P^{-1} = \begin{bmatrix} 2 & 2 \\ -2 & -1 \end{bmatrix}.$$

What does this computation say about the matrix A ?

Task Use the previous computation to determine the eigenvalues of A .

Task Use the previous computation to calculate A^5 .

The previous activity shows why diagonalizing a matrix may be useful. But given a matrix A , how do we potentially find P such that $P^{-1}AP$ is a

diagonal matrix?

Activity

Suppose $P^{-1}AP = D$ for some diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be the columns of P . Note that the equation $P^{-1}AP = D$ holds if and only if $AP = PD$.

Task What happens when I take PD ?

Task Calculate AP and PD in terms of the vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, the scalars $\lambda_1, \dots, \lambda_n$ and the matrix A .

Task So what conditions on P and D are necessary and sufficient for the equation $AP = PD$ to hold?

Theorem An $n \times n$ matrix A is *diagonalizable* if and only if A has n eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ such that the matrix P with columns $\mathbf{v}_1, \dots, \mathbf{v}_n$ is invertible. In this case, $P^{-1}AP = D$, where D is a diagonal matrix such that the elements of D on the main diagonal are the *eigenvalues* of A .

Remark Recalling Invertibility Criteria, the matrix P in Theorem thm-7-1-diagonalizability is invertible if and only if the vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ form a basis for $\mathbb{R}^n$.

If $\mathbb{R}^n$ has a basis of eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$, then every vector $\mathbf{v} \in \mathbb{R}^n$ can be written as a linear combination of these eigenvectors, i.e. there exists constants $a_1, \dots, a_n$ such that

$$\mathbf{v} = a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n.$$

Thus

$$\begin{aligned} T_A(\mathbf{v}) &= A\mathbf{v} \\ &= A(a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n) \\ &= a_1 A\mathbf{v}_1 + a_2 A\mathbf{v}_2 + \dots + a_n A\mathbf{v}_n \\ &= a_1 \lambda_1 \mathbf{v}_1 + a_2 \lambda_2 \mathbf{v}_2 + \dots + a_n \lambda_n \mathbf{v}_n \\ &= \lambda_1(a_1 \mathbf{v}_1) + \lambda_2(a_2 \mathbf{v}_2) + \dots + \lambda_n(a_n \mathbf{v}_n) \end{aligned}$$

In other words, T_A ‘stretches’ the components of $\mathbf{v}$ along each eigenvector by the corresponding eigenvalue.

Activity

Consider the matrix

$$M = \begin{bmatrix} 5/3 & -2/3 \\ -1/3 & 4/3 \end{bmatrix}.$$

Verify that

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}$$

are eigenvectors of M corresponding to different eigenvalues. What are the eigenvalues?

textbook figure fig-7-1-eigenvector-transformation illustrates the effect of the matrix M from Activity ex-eigenvector-verification on a grid in $\mathbb{R}^2$. In particular, it highlights an important geometric interpretation of eigenvectors and eigenvalues; when we apply the linear map defined by the matrix M to any vector in $\mathbb{R}^2$, the matrix map stretches eigenvector directions by their eigenvalues.

Remark: Non-diagonalizable matrices Consider the rotation matrix

$$A_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

where θ is not a multiple of 180° . For any nonzero vector $\mathbf{v} \in \mathbb{R}^2$, the vector $A_\theta \mathbf{v}$ is obtained by rotating $\mathbf{v}$ counter-clockwise by an angle θ , and so $A\mathbf{v}$ is *never* on the line spanned by $\mathbf{v}$, i.e. it is never true that the equation $A\mathbf{v} = \lambda\mathbf{v}$ for any λ , and any non-zero $\mathbf{v}$.

Thus the matrix A_θ has no eigenvectors or eigenvalues. It therefore is a matrix that is not diagonalizable. This makes geometric sense: we cannot find special directions in $\mathbb{R}^2$ so that the action of rotating vectors in the plane is obtained by stretching in those special directions.

Non-diagonalizable matrices illustrates why not every matrix is diagonalizable. A matrix is diagonalizable precisely when it has a basis of eigenvectors, but this is not always possible.

Activity

Let

$$A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}.$$

Is A diagonalizable?

Activity

Let

$$A = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}.$$

Is A diagonalizable?